



PAN-ATLANTIC
UNIVERSITY

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School of Science and Technology

COURSE CODE: DTS304 (BOTH STREAMS)

COURSE TITLE: Database Management I

EXAMINATION: Project (Miniapp powered by Python & PostgreSQL)

INSTRUCTION: Answer all questions

DURATION: 11 DAYS

Section 1: Database (All your SQL codes will be demanded, see section 3)

1. Create a copy of the database **paufs** via an SQL command, name this database “project”.
 - a. Add a new table called **admins** with the following attributes: **adminid**, **adminname**, and **password** (you don’t need to hash it), with the following data types: **int**, **varchar**, and **varchar**, respectively.
 - b. Populate the admins table with three different admin records.
2. Delete all records across all tables.
3. Extend the **Meals** table by adding a new column called **unit_price** (numeric).
4. Extend the **Customer** table, add two new columns called **username** and **password**, they should both be of the varchar data type (**username** should be at least 10 while **password** length should be at least 8).
5. Populate the meals table with at least 12 records.
6. Populate the customer table with at least 15 records.

Section 2: Streamlit and Python (App)

1. Create a Login interface with two tabs:
 - a. The first tab for admins, upon successful login, **redirects** the user to the **admin page**. The admin is required to input his/her **id** and **password** before clicking the login button.
 - b. The second tab for customers, upon a successful login, **redirects** the user to the **customer page**. The customer is required to input his/her **username** and **password** before clicking the login button.
2. In the admin page, there should be **four (4)** tabs:

- a. A tab to update some details of a meal
 - b. A tab to **view recent orders** (List orders from the last x days with customer names. You would notice x is an input from the admin.
 - c. A tab to **view the customers (ids and names) and the average rating scores they placed on orders.**
 - d. A tab to view the **level of customers' activities:** **Count orders per customer** and **compute the total amount spent per customer.**
3. In the customer page, there should be **two (2)** tabs:
 - a. A tab to enable a user (customer) to place a new order. Design this so that an order can feature multiple meals (especially the quantities to order per meal). **Upon a successful order, push the corresponding data into both orders and orderdetails tables.**
 - b. A tab to enable a customer to rate and provide a comment for an order (if s/he wishes to do that). Make sure all the **unrated orders** for the user in session are loaded, so that the user can place the rating score and comment (optional activity) for a selected order.

Section 3: Deliverables

1. An SQL file containing all the activities of **Section 1.**
2. A Python file or multiple Python files (depending on your design choice) containing the Python code written to handle **Section 2.**
3. A Microsoft Word document containing the following:
 - a. Your details (name and matriculation number).
 - b. The exact contents of Deliverable 2 (create sections if you have more than one Python file).
 - c. An explanation of three major aspects of your work (choose any three: could be the section where unrated orders were loaded, or where new orders were inserted into the table, or any other activity you deemed worthy of explanation).